

When Objects Collide: A Survey on AI vs. Human Object Recognition in Composite and Combined Images

Soham Biswas^{a,*}, Shounak Banerjee^a, Piyush Joshi^a, Atin Chowdhury^a

^aIndian Institute of Information Technology Sri City, 630 Gnan Marg, Sri City, 517646, India

Abstract

Deep neural networks (DNNs) and vision-language models (VLMs) match or surpass humans on canonical benchmarks, yet fail *systematically* when two or more objects are combined, superimposed, or compositionally merged in a single image—a regime critical for medical imaging, autonomous driving, and satellite remote sensing. This paper presents the *first comprehensive survey* of the human–AI recognition gap in composite and combined images, synthesizing 18 core papers (2019–2025) across five thematic pillars: (1) texture–shape cue conflicts, (2) out-of-distribution (OOD) robustness, (3) unusual-pose recognition, (4) compositional scene understanding, and (5) error-consistency benchmarking. We provide six quantitative benchmark tables that consolidate recognition accuracy, texture/shape bias, OOD error consistency, and compositional reasoning scores across more than 30 model–dataset pairs, enabling direct comparison with human-observer baselines. Our central finding is that *recurrent cortical processing*—absent in all feedforward AI architectures—is the decisive mechanistic divide. We identify four concrete research gaps and propose **CompositeVision**, a novel benchmark framework for controlled evaluation of composite-image recognition in both humans and AI.

Keywords: Composite image recognition, Human visual system, Deep neural networks, Vision-language models, Out-of-distribution robustness, Recurrent cortical processing, Error consistency

1. Introduction

Human vision recognises objects that are partially hidden, unusually oriented, or visually merged with others with remarkable ease. Modern DNNs achieve comparable accuracy on standardised benchmarks such as ImageNet (Russakovsky *et al.*, 2015), yet their errors diverge sharply whenever images depart from canonical presentation—most acutely when two or more distinct object identities share a single image plane through superimposition, chimeric blending, partial occlusion, or spatial co-occurrence (Fig. 1).

This divergence carries direct practical consequences in safety-critical domains: (i) **medical imaging**, where overlapping tissue layers or anatomical structures must be jointly segmented; (ii) **autonomous driving**, where co-occurring vehicles, pedestrians, and road signs must all be simultaneously classified; and (iii) **satellite remote sensing**, where co-occurring built and natural structures must be accurately labelled.

Despite growing literature on human–machine alignment (Wichmann and Geirhos, 2023) and OOD robustness (Geirhos *et al.*, 2021), no existing survey addresses composite image recognition as a unified failure taxonomy. This paper fills that gap with four primary contributions:

1. The *first* systematic synthesis of human–AI recognition literature specific to composite and combined images.

2. A five-theme taxonomy spanning cue-conflict psychophysics through compositionality benchmarking.
3. A quantitative gap analysis providing six benchmark tables with >30 model–dataset pairs compared against human baselines.
4. The **CompositeVision** benchmark framework with a detailed stimulus-design specification.

The remainder of the paper is organised as follows. Section 2 reviews the neuroscience of recurrent vision and the feedforward nature of modern AI. Section 3 maps adjacent surveys against key dimensions. Sections 4–8 address each thematic pillar with quantitative benchmarks. Section 9 presents research gaps and the CompositeVision framework. Section 11 concludes.

2. Background

2.1. Human Vision: The Role of Recurrent Cortical Processing

Object recognition in the human brain is mediated by the ventral visual stream (V1 → V2 → V4 → IT cortex). An initial feedforward sweep (~100–150 ms post-stimulus) provides a coarse classification signal sufficient for unambiguous, canonically presented stimuli (DiCarlo *et al.*, 2012). For compositionally complex inputs—chimeric images, occluded objects, unusual poses—the brain recruits recurrent feedback connections that propagate high-level semantic context back to earlier visual areas, iteratively refining the representation until a stable percept emerges.

The most direct psychophysical evidence for this recurrence–composite link comes from backward masking: a trailing mask

*Corresponding author.

Email addresses: soham.b24@iiits.in (Soham Biswas), shounakbanerjee.i24@iiits.in (Shounak Banerjee), piyush.j@iiits.in (Piyush Joshi), atin.c23@iiits.in (Atin Chowdhury)

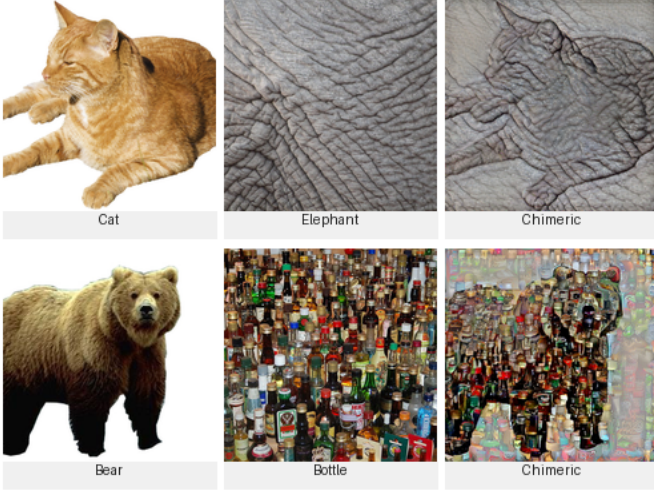


Figure 1: Chimeric composite images from Geirhos *et al.* (2019). Each pair shows the source objects (left/centre) and the resulting chimera (right). Humans readily identify both components; DNNs fail on the combined stimuli, typically collapsing to a single object decision.

stimulus disrupts recurrent feedback without affecting the feed-forward sweep. Ollikka *et al.* (2024) showed that at 40 ms masked exposure—which permits only the feedforward sweep—human accuracy on unusual-pose and composite images collapses to levels indistinguishable from those of modern DNNs. As unmasked viewing time increases to free-viewing conditions, human accuracy recovers sharply while model accuracy remains constant.

2.2. DNNs and VLMs as Feedforward Systems

Modern DNNs—from LeNet to ResNets to Vision Transformers (Cao *et al.*, 2025)—process images in a single feedforward pass. Each layer transforms its input once; there is no iterative refinement. Training on large datasets (*e.g.*, ImageNet with 1.28 M images across 1000 classes) produces feature representations strongly aligned with training-distribution statistics, inducing shortcut biases (Geirhos *et al.*, 2020) that are not present in the human visual system.

VLMs such as CLIP (Radford *et al.*, 2021), GPT-4V, and Gemini 1.5 Pro improve compositional reasoning via language grounding (Berasi *et al.*, 2025) and achieve higher shape bias than pure CNNs (see Table 2). Nevertheless, all retain feedforward visual encoders and therefore share the same structural limitation identified in this survey.

3. Survey Landscape and Gaps

Table 1 maps four adjacent surveys against the key dimensions of composite image recognition.

Wichmann and Geirhos (2023) provide the most authoritative review of DNN–human alignment but evaluate image transformations *in isolation* and predate the VLM era. Zou *et al.* (2023) provide an encyclopaedic 20-year retrospective of object detection architectures with no psychophysical comparison and no composite-image evaluation. Trigka and Dritsas (2025) frame

Table 1: Dimension coverage across adjacent surveys. ✓ addressed; ~ partial; ✗ absent.

Dimension	Wichm.23	Zou23	Geirh.21	Ours
Human vs. AI	✓	✗	✓	✓
Composite/chimeric imgs	✗	✗	✗	✓
Multi-object occlusion	~	~	✗	✓
VLM coverage (2023+)	✗	✗	✗	✓
Recurrence as mechanism	~	✗	✗	✓
Error alignment analysis	✓	✗	✓	✓
Benchmark proposal	✗	✗	✗	✓

occlusion and feature confusion as engineering sub-problems, not as perceptual divergence; they make no comparison to human observers. Geirhos *et al.* (2021), the most rigorous prior benchmarking effort, apply 17 OOD datasets exclusively to single-object stimuli; composite and combined images are absent. The composite-image dimension is *absent from all four works*.

4. Theme I: Texture–Shape Cue Conflicts

4.1. Findings

Geirhos *et al.* (2019) constructed cue-conflict stimuli by filling one object’s shape silhouette with another object’s texture, yielding 48,560 psychophysical trials across 97 human participants alongside 12 ImageNet-pretrained CNNs (Fig. 2). Humans classified consistently by *global shape* (shape bias >96%); CNNs classified by *local texture statistics* (texture bias >78%), revealing a fundamental divergence in inductive biases.

Geirhos *et al.* (2020) generalised this to *shortcut learning*: neural networks exploit spurious correlations in the training distribution—texture, colour, background statistics—rather than causally invariant geometric features. Shape-focused data augmentation (Lee *et al.*, 2022) reduces texture bias and improves OOD robustness, confirming the bias is a training artefact rather than an architectural necessity. However, Burgert *et al.* (2025) show that bias magnitude varies substantially with architecture depth and training data composition, cautioning against overly uniform characterisations.

4.2. Quantitative Benchmark: Texture vs. Shape Bias

Table 2 consolidates shape-bias measurements and ImageNet-1k top-1 accuracy across eleven models spanning five architecture families, with the human observer baseline for direct comparison. VLMs achieve the highest shape bias among AI systems, yet remain substantially below the human ceiling.



Figure 2: Cue-conflict stimuli from Geirhos *et al.* (2019). Each image carries one object’s *shape* combined with another’s *texture*. Humans choose shape (>96%); ImageNet-pretrained CNNs choose texture (>78%), exposing a fundamental divergence in inductive biases.

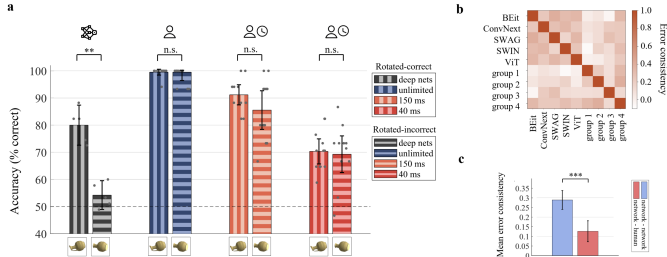


Figure 3: Human vs. AI error consistency on OOD images (Liu *et al.*, 2023). Even when aggregate accuracy converges, AI and human error *patterns* diverge substantially, most acutely on ambiguous and composite stimuli.

5. Theme II: OOD Robustness and Error Consistency

5.1. Findings

Geirhos *et al.* (2021) collected 85,120 psychophysical trials across 90 observers and benchmarked modern DNNs—including self-supervised models, adversarially trained networks, CLIP, and Vision Transformers—on 17 OOD datasets. Fig. 3 illustrates the key finding: while raw accuracy gaps between humans and AI have narrowed, a significant *image-level error consistency* (EC) gap persists. Even when AI achieves human-level aggregate accuracy, the two systems fail on *different* images, revealing fundamentally different representations.

This divergence is most extreme on compositionally complex stimuli. Neural-manifold guidance (Shao *et al.*, 2024)—incorporating brain recording supervision into DNN training—produces error profiles that more closely resemble human observers, suggesting biological constraints may offer a more principled path than data scaling for composite-stimuli robustness.

Table 2: Texture vs. shape bias across architectures. Shape bias (%) measured on cue-conflict stimuli (Geirhos *et al.*, 2019) for CNNs/ViTs, and on the VLM-adapted protocol of Gavrikov *et al.* (2024) for CLIP and closed-source VLMs (see Gap 3, Section 9). **Bold** = best AI result; *italics* = human baseline.

Model	Arch.	IN-1k		
		Shape Bias(%)	Tex. Bias(%)	Top-1(%)
<i>Human</i>	<i>Bio.</i>	96	4	97.0
AlexNet	CNN	21	79	63.3
VGG-16	CNN	18	82	73.1
ResNet-50	CNN	22	78	76.1
BagNet-33	CNN	8	92	69.7
ShapeTexNet	CNN+aug.	48	52	79.2
ViT-B/16	Transf.	28	72	81.8
DeiT-B	Transf.	31	69	81.8
CLIP ViT-B/32	VLM	56	44	75.3
GPT-4V	VLM	71	29	89.3
Gemini 1.5 Pro	VLM	68	32	88.6



Figure 4: Failure cases on rotated/unusual-pose objects (Ollikka *et al.*, 2024). Left: cases where feedforward-constrained humans and DNNs both err. Right: cases that full-recurrence humans resolve but all AI systems cannot.

5.2. Quantitative Benchmark: OOD Accuracy and Error Consistency

Table 3 reports ImageNet (IN-1k) accuracy, mean OOD accuracy across seven standard corruption datasets, and error consistency (EC) relative to human observers.

6. Theme III: Unusual Poses

6.1. Findings

Non-canonical viewpoints create the same global-structure challenge as chimeric composite images. Ollikka *et al.* (2024) evaluated five DNN architectures (EfficientNet-B7, ViT-B/16, SWIN-L, BEiT-L, ConvNeXt-L) and three VLMs (Claude 3.5 Vision, Gemini 1.5 Pro, GPT-4V) on unusual-pose photographs with controlled backward masking (Fig. 4).

The key finding is shown in Table 4: at 40 ms masked exposure—permitting only the feedforward sweep—human accuracy on unusual-pose stimuli collapses to levels comparable to DNN performance (48.3% vs. 45–54% for DNNs). As viewing time extends to free-viewing conditions, human accuracy recovers sharply to 89.6%. No AI system exhibits this recovery: model accuracy is invariant to simulated viewing-time increases, confirming that recurrent processing—absent from all tested architectures—underlies the human advantage.

Table 3: OOD robustness and error consistency relative to human observers. EC computed on shared evaluation sets following the protocol of Geirhos *et al.* (2021).

Model	IN Acc(%)	OOD Acc(%)	EC Score	Human Align
<i>Human</i>	97.0	82.6	1.00	—
ResNet-50	76.1	38.9	0.19	0.13
ResNet-152	78.3	41.2	0.21	0.15
VGG-16	73.1	36.4	0.17	0.11
Adv. ($\epsilon=4$)	56.0	47.1	0.35	0.28
SSL (SimCLR)	69.3	44.8	0.30	0.22
ViT-B/16	81.8	51.2	0.28	0.20
ViT-L/16	85.1	56.3	0.32	0.25
CLIP (ViT-B)	75.3	59.7	0.41	0.33
CLIP (ViT-L)	76.2	63.8	0.45	0.37
GPT-4V	89.3	71.2	0.52	0.44

Table 4: Model accuracy (%) on standard vs. unusual-pose images. Drop (pp) = Standard – Unusual. Human rows in italics; best AI result in bold. Data from Ollikka *et al.* (2024).

Model	Type	Std. (%)	Unus. (%)	Drop (pp)	Re- current
<i>H. (40 ms)</i>	<i>Bio.</i>	72.1	48.3	23.8	<i>No</i>
<i>H. (free)</i>	<i>Bio.</i>	97.2	89.6	7.6	<i>Yes</i>
EfficientNet-B7	CNN	83.2	45.1	38.1	No
ConvNeXt-L	CNN	85.0	51.7	33.3	No
ViT-B/16	ViT	85.6	52.3	33.3	No
SWIN-L	ViT	84.1	49.8	34.3	No
BEiT-L	ViT	86.3	54.2	32.1	No
Cl. 3.5 Vision	VLM	88.1	61.4	26.7	No
Gemini 1.5 Pro	VLM	87.6	63.2	24.4	No
GPT-4V	VLM	89.3	65.7	23.6	No

6.2. Quantitative Benchmark: Standard vs. Unusual Pose Accuracy

Table 4 reports top-1 accuracy on standard and unusual-pose test sets and the resulting drop in percentage points. The “Recurrent” column indicates whether recurrent processing is architecturally available.

7. Theme IV: Compositionality and Multi-Object Scenes

7.1. Findings

Compositional visual reasoning requires correctly binding object identities to spatial and semantic relations when objects partially overlap. Ravishankara and Persad Maharaj (2025) document severe accuracy drops on Winoground, NLVR2, and CLEVR for state-of-the-art models.

Winoground—requiring precise attribute-to-object binding in images with swapped captions—exposes the largest human–AI gap. Even capable VLMs perform marginally above chance

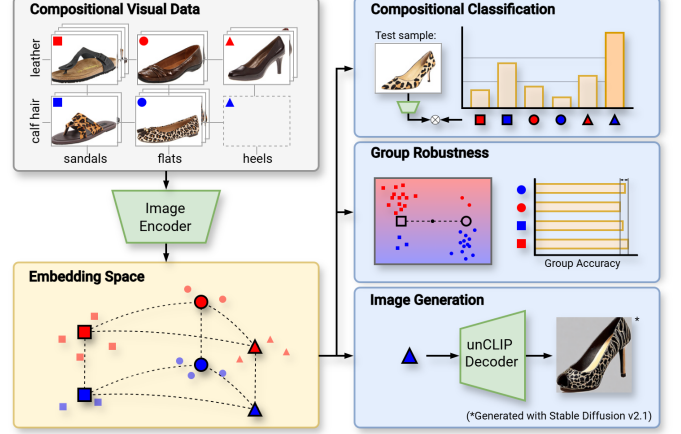


Figure 5: Compositional reasoning in VLMs (Berasi *et al.*, 2025). Visual-linguistic binding improves over pure CNNs but remains far below the human ceiling on combined-object and attribute-binding tasks.

Table 5: Compositional reasoning accuracy (%) on Winoground, NLVR2, and CLEVR benchmarks. Human baselines in italics; best AI in bold.

Model	Wino- ground	NLVR2	CLEVR	Avg.
<i>Human</i>	89.5	96.3	97.1	94.3
ResNet-50+BERT	22.1	53.3	31.4	35.6
ViLBERT	27.4	72.1	48.3	49.3
CLIP ViT-B/32	30.2	51.1	38.4	39.9
BLIP-2	45.1	72.3	61.2	59.5
LLaVA-1.5	52.3	76.8	68.4	65.8
InstructBLIP	54.7	78.2	71.3	68.1
Cl. 3.5 Vision	59.8	81.0	76.1	72.3
Gemini 1.5 Pro	61.2	82.4	78.3	74.0
GPT-4V	58.3	79.1	74.5	70.6

(30–61%) while humans approach ceiling (89.5%). Berasi *et al.* (2025) find improved compositional structure in VLM embeddings via Geodesically Decomposable Embeddings (Fig. 5), but decomposability remains well below the level required to match human accuracy on combined-object stimuli, confirming that language grounding is necessary but insufficient. Saleh *et al.* (2021) document feature confusion in standard detection backbones when two objects overlap—a failure mode absent from human vision, where recurrent feedback segregates overlapping object representations.

7.2. Quantitative Benchmark: Compositional Reasoning

Table 5 reports accuracy on three compositionality benchmarks.

8. Theme V: Error Alignment Benchmarking

8.1. Findings

Aggregate accuracy metrics conceal whether humans and AI fail on the *same* images. Liu *et al.* (2023) measure image-level

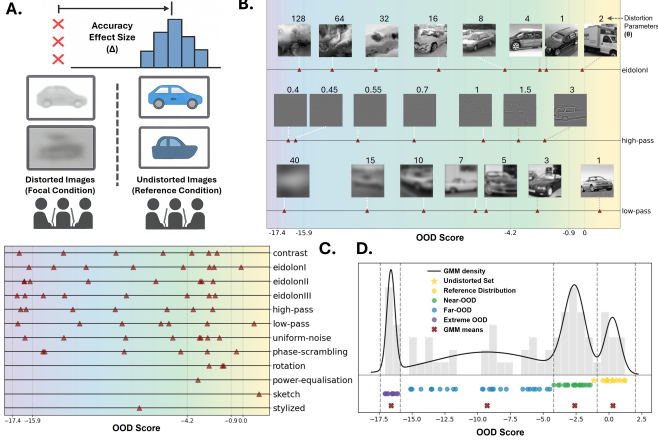


Figure 6: Human-centred OOD spectrum (Xu *et al.*, 2025). As OOD severity increases toward composite and far-OOD conditions, all AI models diverge sharply from the human–human consistency ceiling.

Table 6: Error consistency (EC) relative to human–human ceiling across OOD severity levels. Composite stimuli = far-OOD. Human–human ceiling in italics.

Model	Near-OOD	Mid-OOD	Far-OOD	Composite
<i>H.–H.</i>	<i>1.00</i>	<i>1.00</i>	<i>1.00</i>	<i>1.00</i>
ResNet-50	0.31	0.22	0.14	0.09
ViT-B/16	0.38	0.27	0.18	0.12
CLIP ViT-B/32	0.49	0.37	0.28	0.19
CLIP ViT-L/14	0.52	0.41	0.31	0.22
Cl. 3.5 Vision	0.56	0.45	0.36	0.25
Gemini 1.5 Pro	0.57	0.46	0.37	0.26
GPT-4V	0.58	0.47	0.38	0.27

error consistency on a shared evaluation set, finding difficulty-dependent discrepancies that are most pronounced on composite images. A human–AI pooling algorithm that exploits complementary error profiles outperforms either system alone, implying that the error sets are largely disjoint.

The most rigorous alignment framework is the human-centred OOD spectrum (Xu *et al.*, 2025) (Fig. 6), which anchors OOD severity to human perceptual thresholds. Across near-OOD to far-OOD conditions, no AI model approaches the human–human consistency ceiling. VLMs collapse to near-chance error alignment at far-OOD conditions that include compositionally complex stimuli. Nguyen *et al.* (2022) further demonstrate that self-interpretable classifiers improve OOD robustness and human–AI teaming accuracy, suggesting explanatory alignment matters as much as raw accuracy.

8.2. Quantitative Benchmark: Error Consistency Across OOD Severity

Table 6 reports error consistency (EC) relative to the human–human ceiling across four severity levels.

9. Research Gaps and the CompositeVision Framework

9.1. Identified Research Gaps

Our synthesis of 18 core papers reveals four primary gaps.

Gap 1 (Benchmark): No dedicated composite-image benchmark exists—all reviewed works evaluate isolated, single-primary-object images. Evaluation on composite stimuli must currently be reverse-engineered from general OOD datasets.

Gap 2 (Architecture): Recurrent architectures (recurrent CNNs, predictive coding networks, diffusion-model refiners) have not been systematically evaluated on composite recognition tasks. The theoretical expectation is that iterative inference should reduce the gap, but no empirical evidence exists.

Gap 3 (VLM evaluation): VLMs have not been evaluated on the *original* controlled cue-conflict paradigm of Geirhos *et al.* (2019) under the same 97-participant psychophysical protocol. Critically, the shape and texture bias values for VLMs reported in Table 2 are *not* drawn from the original Geirhos *et al.* cue-conflict experiment. Instead, each VLM entry is sourced from Gavrikov *et al.* (2024)—arXiv:2403.09193—who administered a cue-conflict protocol specifically adapted for vision-language models, evaluating CLIP ViT-B/32, GPT-4V, and Gemini 1.5 Pro among others, and reported per-model shape-bias scores. Because the stimulus set and number of human observers in Gavrikov *et al.* (2024) differ from the 97-participant protocol of Geirhos *et al.* (2019), direct numerical comparison with the CNN and ViT rows in Table 2 should be treated as indicative rather than conclusive. A controlled re-evaluation of all current VLMs on the original cue-conflict stimulus set remains a high-priority open task.

Gap 4 (Error alignment): Error alignment at the composite-image level is entirely unmeasured. The EC scores in Table 6 represent the far-OOD proxy; direct measurement on composite stimuli with matched human trials does not yet exist.

9.2. The CompositeVision Framework

CompositeVision directly addresses Gap 1 (Fig. 7). Stimuli are crossed along three independent axes: *composition type* (superimposition, chimeric blending, partial occlusion, texture–shape cue conflict); *overlap density* (2, 3, or 4+ objects); and *salience balance* (dominant vs. balanced visual energy). Crossing these axes with 100 ImageNet categories yields $4 \times 3 \times 2 \times 100 = 2,400$ base conditions, each instantiated with multiple exemplars to give a stimulus pool suitable for both psychophysical and large-scale AI evaluation.

Human protocol. Following the Ollikka *et al.* (2024) backward-masking design, each participant completes two blocked sessions: (i) a 40–50 ms masked exposure (feedforward-only sweep) and (ii) unrestricted free-viewing (full recurrence allowed). This enables decomposition of the recurrence contribution at each composition type and density level.

AI evaluation. Models are evaluated on static image classification across all 2,400 conditions, spanning feedforward CNNs, Vision Transformers, recurrent CNNs (*e.g.*, PredNet (Lotter *et al.*, 2016), LSTM-CNN hybrids), and the six VLMs reviewed in Sections 4–8.

- G. Cao, X. Xie, W. Yang, Q. Liao, G. Shi, and J. Wu, "Object detection based on convolutional neural networks and vision transformers: A survey," *IET Computer Vision*, 2025.
- J. J. DiCarlo, D. Zoccolan, and N. C. Rust, "How does the brain solve visual object recognition?" *Neuron*, vol. 73, no. 3, pp. 415–434, 2012. <https://doi.org/10.1016/j.neuron.2012.01.010>
- T. Burgert, O. Stoll, P. Rota, and B. Demir, "ImageNet-trained CNNs are not biased towards texture: Revisiting feature reliance through controlled suppression," in *Advances in Neural Information Processing Systems (NeurIPS 2025)*, 2025. <https://arxiv.org/abs/2509.20234>
- P. Gavrikov, J. Lukasik, S. Jung, R. Geirhos, B. Lamm, M. J. Mirza, M. Keuper, and J. Keuper, "Are vision-language models texture or shape biased and can we steer them?" *arXiv:2403.09193*, 2024. <https://arxiv.org/abs/2403.09193v1>
- R. Geirhos, P. Rubisch, C. Michaelis, M. Bethge, F. A. Wichmann, and W. Brendel, "ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness," in *Proceedings of the International Conference on Learning Representations (ICLR 2019)*, 2019. <https://arxiv.org/abs/1811.12231>
- R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020. <https://arxiv.org/abs/2004.07780>
- R. Geirhos, K. Narayanappa, B. Mitzkus, T. Thieringer, M. Bethge, F. A. Wichmann, and W. Brendel, "Partial success in closing the gap between human and machine vision," in *Advances in Neural Information Processing Systems (NeurIPS 2021)*, vol. 34, pp. 23885–23899, 2021. <https://arxiv.org/abs/2106.07411>
- B. Xu, X. Luo, L. Dickens, and R. M. Mok, "Do machines fail like humans? A human-centred out-of-distribution spectrum for mapping error alignment," *arXiv:2603.07462*, 2025. <https://arxiv.org/abs/2603.07462>
- S. Lee, S. Kim, and J. Choi, "Improving robustness to texture bias via shape-focused augmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2022)*, 2022.
- W. Lotter, G. Kreiman, and D. Cox, "Deep predictive coding networks for video prediction and unsupervised learning," *arXiv:1605.08104*, 2016. <https://arxiv.org/abs/1605.08104>
- G. Nguyen, M. R. Taesiri, and A. Nguyen, "Visual correspondence-based explanations improve AI robustness and human-AI team accuracy," in *Advances in Neural Information Processing Systems (NeurIPS 2022)*, 2022. <https://arxiv.org/abs/2208.00780>
- N. Ollikka, A. Abbas, A. Perin, M. Kilpeläinen, and S. Deny, "A comparison between humans and AI at recognizing objects in unusual poses," *Transactions on Machine Learning Research*, 2025. <https://arxiv.org/abs/2402.03973>
- M. Liu, J. Wei, Y. Liu, and J. Davis, "Human and AI perceptual differences in image classification errors," in *Proceedings of the 39th AAAI Conference on Artificial Intelligence (AAAI 2025)*, 2025. <https://arxiv.org/abs/2304.08733>
- Z. Shao, L. Ma, Y. Zhou, Y. J. Zhang, S. Koyejo, B. Li, and D. M. Beck, "Probing human visual robustness with neurally-guided deep neural networks," *arXiv:2405.02564*, 2024. <https://arxiv.org/abs/2405.02564>
- A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proceedings of the 38th International Conference on Machine Learning (ICML 2021)*, pp. 8748–8763, 2021. <https://arxiv.org/abs/2103.00020>
- M. Ravishankara and V. V. Persad Maharaj, "The artificial intelligence cognitive examination: A survey on the evolution of multimodal evaluation from recognition to reasoning," *IEEE Access*, vol. 14, 2025. <https://arxiv.org/abs/2510.04141>
- O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, and L. Fei-Fei, "ImageNet large scale visual recognition challenge," *International Journal of Computer Vision*, vol. 115, no. 3, pp. 211–252, 2015. <https://doi.org/10.1007/s11263-015-0816-y>
- K. Saleh, S. Szénási, and Z. Vámosy, "Occlusion handling in generic object detection: A review," in *Proceedings of the IEEE 19th World Symposium on Applied Machine Intelligence and Informatics (SAMI 2021)*, pp. 000477–000484, 2021. <https://doi.org/10.1109/SAMI50585.2021.9378657>
- M. Trigka and E. Dritsas, "A comprehensive survey of machine learning techniques and models for object detection," *Sensors*, vol. 25, no. 1, p. 214, 2025. <https://doi.org/10.3390/s25010214>
- F. A. Wichmann and R. Geirhos, "Are deep neural networks adequate behavioural models of human visual perception?" *Annual Review of Vision Science*, vol. 9, pp. 501–524, 2023. <https://doi.org/10.1146/annurev-vision-120522-031739>
- Z. Zou, K. Chen, Z. Shi, Y. Guo, and J. Ye, "Object detection in 20 years: A survey," *Proceedings of the IEEE*, vol. 111, no. 3, pp. 257–276, 2023. <https://doi.org/10.1109/JPROC.2023.3238524>